DIGI Node — Tier 1: Fully Assembled Unit

Built, soldered and bench-tested. Meshtastic is flashed on the radio, and the radio's serial link to the node is already configured.

In the box

Item	Notes
Assembled node	8 GB microSD fitted, TPU seals fitted
Active GPS helix antenna	Center SMA
2.4 GHz Wi-Fi/BLE SMA antenna	Left SMA
LoRa SMA antenna for your region (915 MHz US)	Right SMA

Not included: two 18650 batteries. Use flat-top cells of the same brand and capacity.

Set it up

1. Remove the back cover and fit the two batteries in the correct orientation.
2. Screw on the antennas: Wi-Fi on the left, GPS in the center, LoRa on the right. Tighten finger tight, then another eighth of a turn.
3. Flash AntiHunter on the ESP32 over USB with the web flasher at lukeswitz.github.io/AntiHunter. Choose **Full** for the web UI or **Headless** for quiet field operation.
4. Set the LoRa region on the radio. The region is **UNSET** when the unit leaves the bench. Until you set it the radio only receives, and the node will not answer over mesh.
5. Change the radio's Bluetooth pairing PIN, because every unit leaves the bench with the same one. For quieter operation you can turn Bluetooth off and manage the Heltec board over a cable.
6. Make your own encrypted Meshtastic channel the primary channel and turn the public default channel off.
7. **Full firmware only:** join the `Antihunter` Wi-Fi network (password `antihunt3r123`), open `http://192.168.4.1` and change these credentials before doing anything else.
8. Set an erase key. Erase commands are refused until you do. Full: System tab → **Secure Data Destruction**. Headless: the **Erase key** field in the web flasher.
9. Fit the USB seal before using the node outdoors. The GPS antenna has its own rubber seal, so you only need the separate antenna seal if you change antennas. The enclosure is weather-resistant, not waterproof. Shade the node from direct sun where you can.
10. Watch the serial log on first boot. Any hardware problem shows up there. Shipping can loosen the SD card, so if you see SD errors, make sure it is fully seated.

> **Never power the node without the LoRa antenna attached.**

Power

- Use a regulated 5 V supply only, from USB or the UPS board. Never wire cells or an unregulated supply to the PCB.
- Connect **one USB cable at a time.**
- **Red LED on the UPS:** a faint glow is normal whenever batteries are fitted. Its draw is negligible.
- **Blue LED on the UPS:** on while charging, and on some batches also while the boost rail is active. It goes off when the batteries are full.
- **Fan:** blows outward and switches on when the thermal switch closes. To run it all the time, short the two `THERMO` pins.

Full guide

For each step above in detail, plus antennas, all detectors, the vibration sensor, mesh commands and Command Center, see the Operator's Guide: github.com/lukeswitz/AntiHunter/blob/main/docs/AntiHunter-Operators-Guide.pdf

- **Firmware, docs, issues:** github.com/lukeswitz/AntiHunter
- **Command Center:** github.com/TheRealSirHaXalot/AntiHunter-Command-Control-PRO